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2023 Trial Examination

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STUDENT NUMBER									Letter

SPECIALIST MATHEMATICS

Written examination 1

Reading time: 15 minutes
Writing time: 1 hour

QUESTION AND ANSWER BOOK

Structure of book

Number of questions	*Number of questions to be answered*	*Number of marks*
9	9	40

- Students are permitted to bring into the examination room: pens, pencils, highlighters, erasers, sharpeners and rulers
- Students are NOT permitted to bring into the examination room: notes of any kind, blank sheets of paper and/or white out liquid/tape.
- No calculator is permitted in this examination.

Materials supplied

- Question and answer book of 11 pages.

Instructions

- Print your name in the space provided on the top of this page.
- All written responses must be in English.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic communication devices into the examination room.

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Instructions

Answer **all** questions in pencil on the answer sheet provided.
Unless otherwise specified, an **exact** answer is required to a question.
In questions where more than one mark is available, appropriate working **must** be shown.
Unless otherwise indicated, the diagrams in this book are not drawn to scale.
Take the acceleration due to gravity to have magnitude $g\ ms^{-2}$, where $g = 9.8$.

Question 1 (6 marks)

Consider the vectors $\underset{\sim}{a} = -\underset{\sim}{i} + c\,\underset{\sim}{j} + 3\,\underset{\sim}{k}$ and $\underset{\sim}{b} = 3\,\underset{\sim}{i} + 6\,\underset{\sim}{j} + d\,\underset{\sim}{k}$ where $c, d \in R$

a. Find the value(s) of c given the magnitude of $\underset{\sim}{a}$ is $\sqrt{14}$.

1 mark

b. If instead, the two vectors are parallel, find the values of c and d.

2 marks

c. Find the cross product of $\underset{\sim}{a} = -\underset{\sim}{\mathrm{i}} + c\,\underset{\sim}{\mathrm{j}} + 3\,\underset{\sim}{\mathrm{k}}$ and $\underset{\sim}{b} = 3\,\underset{\sim}{\mathrm{i}} + 6\,\underset{\sim}{\mathrm{j}} + d\,\underset{\sim}{\mathrm{k}}$ hence find the values of c, d and f, (different to either **part a.** or **part b.**), so that $\underset{\sim}{a}$ and $\underset{\sim}{b}$ are both perpendicular to $\underset{\sim}{e} = -2\,\underset{\sim}{\mathrm{i}} + \underset{\sim}{\mathrm{j}} + f\,\underset{\sim}{\mathrm{k}}$

3 marks

1 + 2 + 3 = 6 marks

TURN OVER

Question 2 (3 marks)

Given $f(x) = x\tan^{-1}\left(\frac{x}{2}\right)$ find $f'(-2)$, writing your answer in the form $m\pi - n$ where m, n are simplified fractions.

3 marks

Question 3 (3 marks)

Prove, by mathematical induction, that $x^2 + x$ is an even integer for $x \epsilon N$.

3 marks

Question 4 (6 marks)

Given $\sec^2(2x) = 2$, where x is in the first quadrant, evaluate:

a. x

b. $\operatorname{cosec}(4x)$

c. $\cot(4x)$

2 + 2 + 2 = 6 marks

TURN OVER

Question 5 (6 marks)

Evaluate the following integrals, leaving your answer in exact, simplified form.

a. $\int_0^1 \left(x \tan^{-1} x\right) \, dx \, dx$

3 marks

b. $\int_0^{\frac{1}{2}} \left(\frac{2x+1}{\sqrt{1-x^2}}\right) dx$

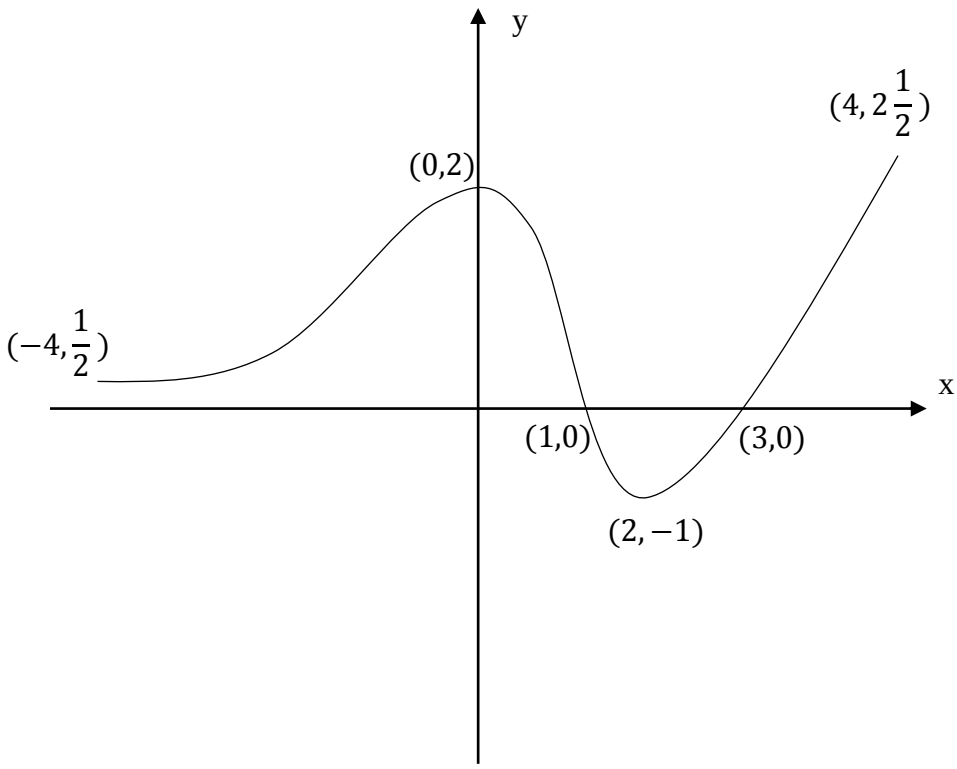
3 marks

3 + 3 = 6 marks

TURN OVER

Question 6 (3 marks)

Sketch on the same axes the graph of $y = \frac{1}{f(x)}$ given the graph of $y = f(x)$ shown below. Label key features of your sketch



3 marks

Question 7 (6 marks)

a. Write $z = -2\sqrt{2} + 2\sqrt{2}i$ in polar form where $z \in C$.

2 marks

b. Hence find two solutions to $z^2 = -2\sqrt{2} + 2\sqrt{2}i$ in polar form.
(Call these solutions z_1 and z_2).

2 marks

c. Use the identity $\cos 2\theta = 2cos^2\theta - 1$ to rewrite the solution to z_1 found in **part b.** into the form $z_1 = \sqrt{a + \sqrt{b}} + \sqrt{a - \sqrt{b}}\, i$, where $a, b, c \in R$

TURN OVER

2 marks

2 + 2 + 2 = 6 marks

Question 8 (4 marks)

Find the gradients of the tangents to $2x^3 - xy^2 = y$ when $x = -1$.

4 marks

Question 9 (3 marks)

The following may be useful for question 9.
Let Z be an observation from a standard normal distribution.
$\Pr(Z \leq 1.96) \approx 0.975$, $\Pr(Z \leq 2.58) \approx 0.995$, $\Pr(Z \leq 3.00) \approx 0.999$

Suppose that the volume of lemonade delivered to a cup by a vending machine (X) is normally distributed with a mean of $200\,mL$ and a standard deviation of $2.5\,mL$. Find the percentage chance that a sample of 25 drinks will produce a mean volume of greater than $201.5mL$.

3 marks

END OF QUESTION AND ASNWER BOOK